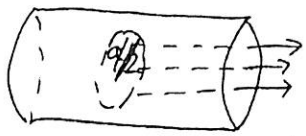
7.12



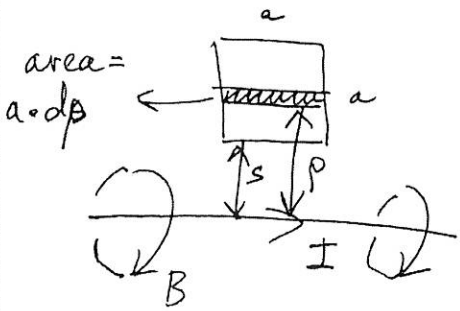
$$B(t) = B_0 \cos(\omega t)\,\hat{z}$$

$$\Phi = \int B(t)\,da = B_0 \cos(\omega t)\,\pi\left(\frac{a}{2}\right)^2$$

$$\mathcal{E}(t) = -\frac{d\Phi}{dt} = \frac{\pi a^2}{4} B_0 \omega \sin(\omega t)$$

$$I(t) = \frac{\mathcal{E}(t)}{R} = \frac{\pi a^2}{4}\frac{B_0 \omega}{R}\sin(\omega t)$$

$\boxed{7.18}$



Current I produces $\vec{B}(s) = \dfrac{\mu_0 I}{2\pi s}\,\hat{\varphi}$

$$\phi = \int \vec{B}\cdot d\vec{A} = \frac{\mu_0 I}{2\pi}\int_s^{s+a} \frac{a\cdot d\rho}{\rho} =$$

$$= \frac{\mu_0 I a}{2\pi}\ln\left(\frac{s+a}{s}\right).$$

<u>Check</u> Area of the loop:

$$\int_s^{s+a} a\,d\rho = a\,\rho\Big|_s^{s+a} = a(s+a-s) = a^2 \text{ (of course)}$$

$$I_{loop} = \frac{1}{R}\,\mathcal{E} = \frac{1}{R}\left(-\frac{d\phi}{dt}\right) = -\frac{\mu_0 a}{2\pi R}\ln\left(\frac{s+a}{s}\right)\cdot\frac{dI(t)}{dt}$$

Flux ϕ is decreasing as $\dfrac{dI}{dt} = \begin{cases} -\alpha I & \text{for } t \le \frac{1}{\alpha} \\ 0 & \text{for } t > \frac{1}{\alpha}. \end{cases}$

$\Downarrow$

I_{loop} will try to support the flux $\Rightarrow$ I_{loop} flows counterclockwise in order to generate field $\vec{B}'$ pointing out of the page.

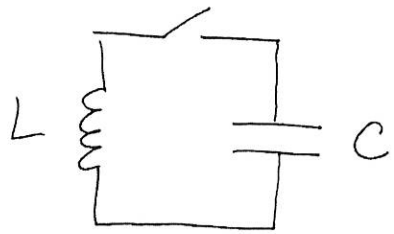
Total charge passed through the loop

$$Q = \int_0^\infty I(t)\,dt = \frac{1}{R}\int_0^\infty\left(-\frac{d\phi}{dt}\right)dt =$$

$$= -\frac{1}{R}\Big(\underbrace{\phi(t=\infty)}_{0} - \underbrace{\phi(t=0)}_{\frac{\mu_0 I a}{2\pi}\ln\left(\frac{s+a}{a}\right)}\Big) = \frac{\mu_0 I a}{2\pi R}\ln\left(\frac{s+a}{s}\right).$$

Note: solution in the <u>manual</u> is not correct! ☺

$\boxed{7.25}$



$Q = CV$ at $t=0$, and $I(t=0)=0$.

$$-L\frac{dI}{dt} = V = \frac{Q}{C}$$

$$I = \frac{dQ}{dt}$$

$$\Rightarrow L\frac{d^2Q}{dt^2} + \frac{Q}{C} = 0$$

$$\frac{d^2Q}{dt^2} = -\frac{1}{LC}Q = -\omega_0^2 Q$$

$$Q(t) = A\cos\omega_0 t + B\sin\omega_0 t$$

$$I(t) = -A\omega_0 \sin\omega t + B\omega_0 \cos\omega_0 t.$$

Initial conditions: $Q(t=0) = CV = A$

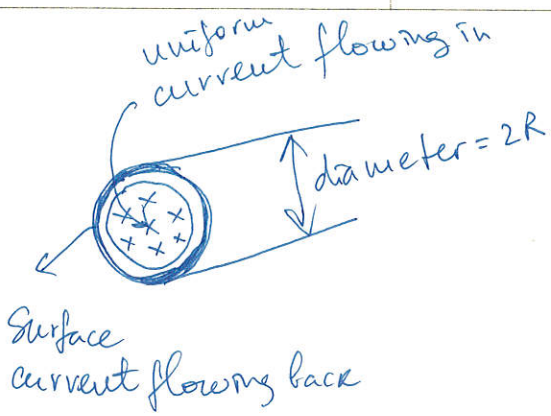
$I(t=0) = B\omega_0 = 0$

$$\Rightarrow Q(t) = CV\cos\left(\frac{t}{\sqrt{LC}}\right)$$

$$I(t) = -\frac{CV}{\sqrt{LC}}\sin\left(\frac{t}{\sqrt{LC}}\right) = -V\sqrt{\frac{C}{L}}\sin\left(\frac{t}{\sqrt{LC}}\right).$$

Resistance R will cause energy dissipation and decay of current with time.

P7.28



Find self-inductance by calculating magnetic energy

$$\oint \vec{B}\cdot d\vec{l} = B\cdot 2\pi r = \mu_0 I_{enc} = \mu_0 I \frac{\pi r^2}{\pi R^2}$$

$$B = \mu_0 I \frac{r}{2\pi R^2}$$

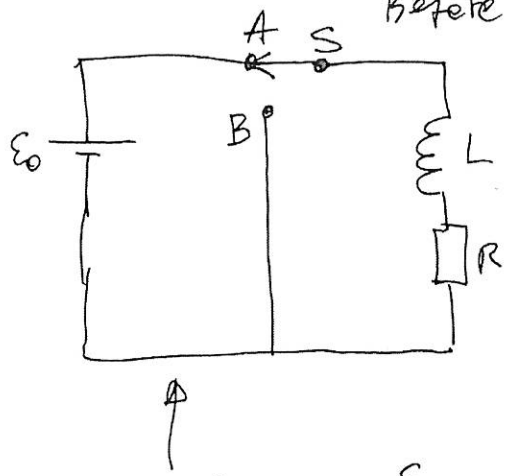
Energy $$W_{mag} = \frac{1}{2\mu_0}\int B^2 d^3r = \frac{1}{2\mu_0}\left(\frac{\mu_0 I}{2\pi R^2}\right)^2 \int r^2 \cdot 2\pi r\,dr\,\underbrace{dz}_{\text{integration along length (axis)}}$$

$$= \frac{1}{2\mu_0}\frac{\mu_0^2 I^2}{2\pi R^4}\underbrace{\int_0^R r^3 dr}_{\frac{1}{4}R^4}\cdot \underbrace{l}_{\text{length}} = \frac{\mu_0 I^2}{16\pi} l$$

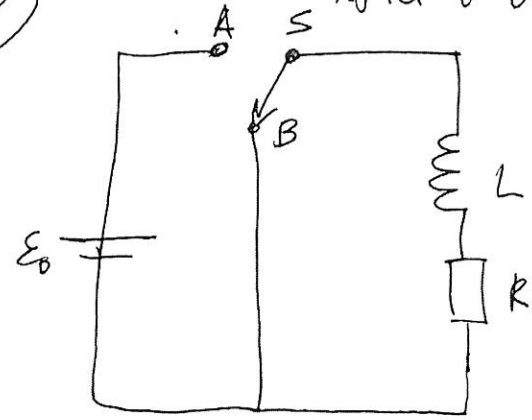
But $W_{mag} = \frac{1}{2}\underbrace{L}_{\text{inductance}} I^2 \Rightarrow L_{ind} = \frac{\mu_0}{8\pi} l \Rightarrow \underbrace{\frac{L_{ind}}{l}}_{\text{inductance per unit length}} = \frac{\mu_0}{8\pi}$.

7.29

Before $t=0$



After $t=0$:



steady current $I_0 = \frac{\mathcal{E}_0}{R}$

(a) At $t \geq 0$: battery is disconnected $\Rightarrow$ emf $= -L\frac{dI}{dt}$ only.

$$-L\frac{dI}{dt} = RI \Rightarrow \frac{dI}{dt} = -\frac{R}{L}I$$

$$I(t) = I_0 e^{-\frac{R}{L}t} = \frac{\mathcal{E}_0}{R} e^{-\frac{R}{L}t}$$

(b) $P = \mathcal{E} I = -L\frac{dI}{dt} I = -\frac{1}{2}L\frac{dI^2}{dt}$

Total energy $W = \int P\,dt = -\frac{1}{2}L\int_0^\infty \frac{dI^2}{dt}dt = \frac{1}{2}L I_0^2$

$$W = \frac{1}{2}L\left(\frac{\mathcal{E}_0}{R}\right)^2$$

(c) Initial energy of the inductor $= \frac{1}{2}L I_0^2$ by def.

$\Rightarrow$ coincides with W in (b), of course.